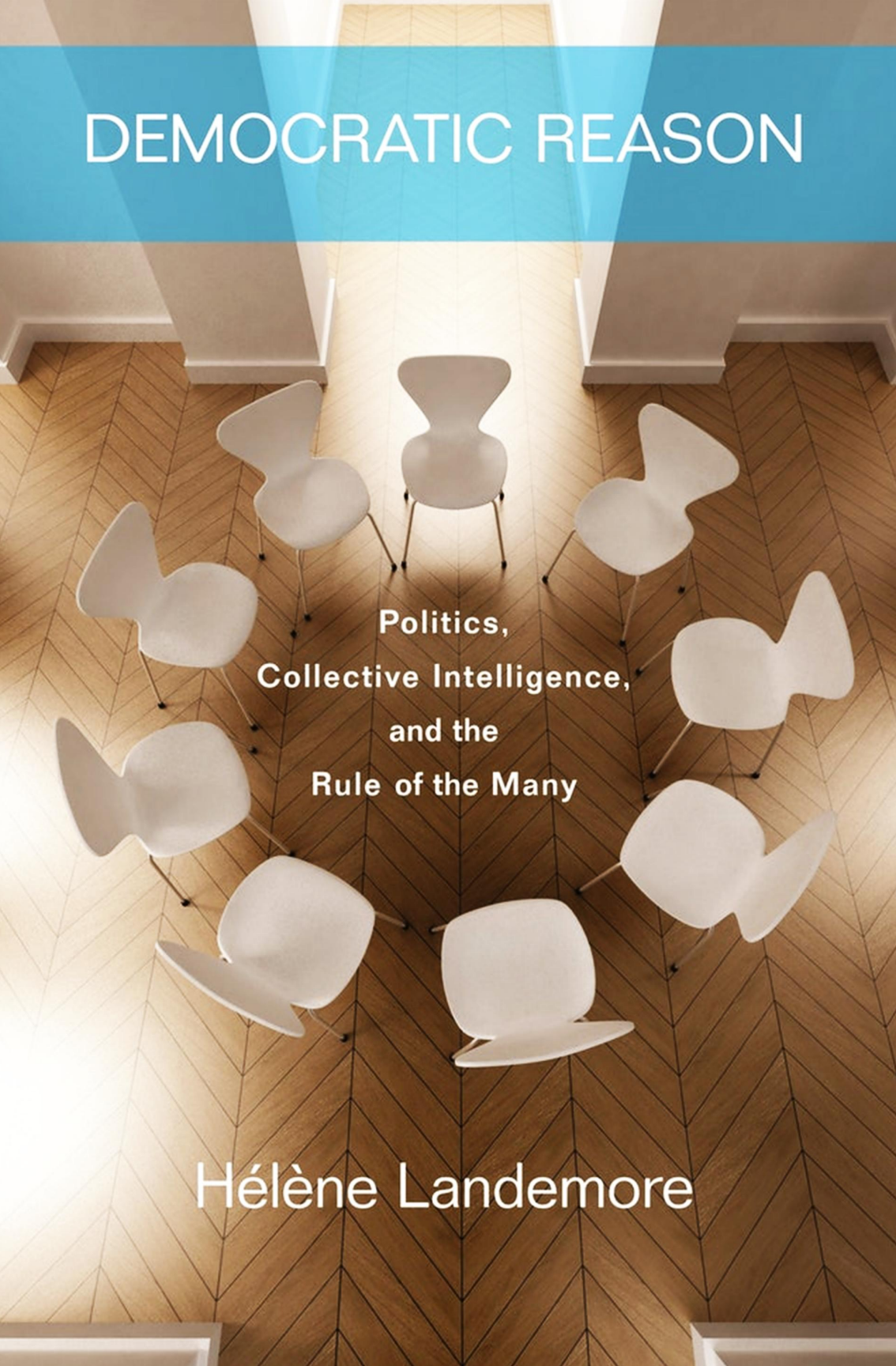


DEMOCRATIC REASON



Politics,
Collective Intelligence,
and the
Rule of the Many

Hélène Landemore

Democratic Reason

POLITICS, COLLECTIVE INTELLIGENCE,
AND THE RULE OF THE MANY

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The Maze and the Masses

DEMOCRACY IS GENERALLY HAILED, in the West at least,¹ as the only legitimate form of government. We (Westerners) only consider legitimate those regimes that are democratic or in the process of becoming so. Conversely, anything undemocratic raises our suspicion. In fact, democracy has such positive valence that some have argued that it is more than a descriptive term objectively referring to a certain type of regime or system of government; in this account, it is an evaluative term, by which its users commend certain institutions and societies (e.g., Skinner 1973). Even if we take democracy to simply denote a certain type of rule characterized by popular participation, it is a fact that this rule has, in the Western world, a privileged aura of legitimacy that competing rules are lacking. At the same time, however, there exists among contemporary democratic theorists, and even among the people themselves, a widely shared skepticism about the capacity of the people for self-rule.

The idea of democracy as a competent regime, in the sense of intelligent or even wise, is not intuitive. Ironically, indeed, democracy could easily be construed as the rule of “the idiots” (the Greek *hoi idiotai* stand for “the ordinary citizens”). In the philosophy of thought, political intelligence is generally a quality attributed to aristocracies, monarchies, and other elitist regimes—not democracies. If democracy is valuable, most people would argue, it is for reasons that have nothing to do with the intelligence, let alone the wisdom, of the masses. Indeed, these reasons have to override the notorious fact of the “folly of crowds.”

Contra this widely shared intuition that democracy is a right that the people do not really possess the competence to exercise, this book consists in a sustained epistemic argument for democracy based on the idea of collective intelligence. I argue that democracy is a smart collective decision-making procedure that taps into the intelligence of the people as a group in ways that can even, under the right conditions, make it smarter than alternative regimes such as the rule of one or the rule of the

¹In the following I consider my natural audience to be Western (or at least Westernized) political theorists and citizens in general. My excuse for such a narrow conception of the public worth addressing is that I would never have gotten started if I had to account for all the philosophical assumptions and principles underpinning the argument of this book.

few. This idea of collective intelligence offers, in theory, an attractive solution to the problem of the average citizen's ignorance and irrationality. If the many as a group can be smarter than any individual within them, then political scientists need not worry so much about the cognitive performance of the average voter and should focus instead on the emergent cognitive properties of the people as a group.

The argument, like many arguments in political theory, is of course not entirely new and has in fact a decent philosophical pedigree. It was first considered, with skepticism, by Aristotle (who himself borrowed it from the Sophists) and, as I will have the leisure to show at some length later on, has been running ever since in different guises as an underground current of political philosophy in the mainstream suspicion toward democracy. In contemporary democratic theory, as I will review in chapter 2, different versions and parts of the argument have been recently taken up by both political scientists and normative political philosophers seeking to use them for the justification of democracy.

This book offers an overview and assessment of the arguments that can be advanced in favor of what I call “democratic reason” or the collective intelligence of the people. I use these arguments, in connection to a literature on the wisdom of crowds, to support a strong epistemic case for democracy. I also develop the argument in a comparative manner, contrasting the epistemic benefits of democracy with those of non-democratic decision procedures. Although I proceed essentially from an a priori and theoretical perspective, reasoning about democracy as an ideal type and a model, the argument that democracy is an epistemically superior form of decision making can be translated into an empirical claim that lends itself to falsification and can be supported by empirical evidence. The kind of empirical evidence that would tend to support the strong epistemic claim I put forward can be illustrated by Josiah Ober's recent study of Athenian democracy (e.g., Ober 2010 and 2012), which establishes that the superiority of classical Athens over rival city-states was due to its ability to process the distributed knowledge and information of its citizens better than less democratic regimes.² The institutions

²I became aware of Ober's manuscript *Democracy and Knowledge* when it was still unpublished but too late in the process of writing my own book to integrate its content properly. Ober later kindly accepted an invitation to write an essay for the edited volume *Collective Wisdom* that Jon Elster and I had been putting together since 2008 (now published with Cambridge University Press in 2012). His work provides, in my view, a compelling historical illustration, if not a systematic demonstration per se, of both Page's (2007) thesis that cognitive diversity matters to collective intelligence and my conjecture that under some reasonable assumptions and all things being equal otherwise, democracy is epistemically smarter than the rule of the few.

accounting for the epistemic superiority of Athenian democracy include, for example, the Council of 500 and the practice of ostracism.

1. THE MAZE AND THE MASSES

The heart of the book is thus a defense of the “collective intelligence” hypothesis in favor of democracy. I argue that there are good reasons to think that for most political problems and under conditions conducive to proper deliberation and proper use of majority rule, a democratic decision procedure is likely to be a better decision procedure than any nondemocratic decision procedures, such as a council of experts or a benevolent dictator. I thus defend a strong version of the epistemic argument for democracy.³ In my view, all things being equal otherwise, the rule of the many is at least as good as, and occasionally better than, the rule of the few at identifying the common good and providing solutions to collective problems. This is so, I will suggest, because including more people in the decision-making process naturally tends to increase what has been shown to be a key ingredient of collective intelligence in the contexts of both problem solving and prediction—namely, cognitive diversity.

I will explain at length what I mean by cognitive diversity later, but let me for now illustrate the epistemic argument for democracy presented here with a metaphor, inspired by Descartes’s thought experiment in the *Discourse on Method*,⁴ which, for all its limitations, should prove enlightening.

Imagine a maze in which a group of people happens to be lost. This maze has an exit (perhaps even several). Clues as to the way of finding the exit(s) are written on the walls. The clues are written in different languages and sometimes coded in pictograms and equations. The clues are dispersed all over the maze, sometimes inscribed very high on the wall, sometimes very close to the ground. Some are written in small fonts, some in very large fonts. The group itself is a typical sample of humanity. It is thus composed of very different people. Some people are nearsighted and some are farsighted. Some are good at mathematics and some are good at languages. Some are very bright and some are not bright at all. The members of the group care about each other and they have only one good: to get out of the

³The weak version of the epistemic argument for democracy holds that democracies are only better than random at making decisions.

⁴In the interest of full disclosure, I admit it is also inspired by a less highbrow source: a Canadian horror movie entitled *Cube* (1997), involving a group of people with different cognitive skills who learn (the hard way of course) that their only chance of survival in the “cube” is to collectively solve riddles, which requires cooperation and a pooling of their respective talents.

maze, preferably together. Every time they reach a fork in the maze,⁵ they have to make a decision as to which direction the group should take. What kind of decision procedure should the group commit to at the beginning if their goal is to maximize their chances to get out of the maze?

Let us assume for now that the commitment is final and that the decision rule cannot be renegotiated at every fork in the road. At the beginning of the journey, the group faces the following set of choices. A first option is for the group to flip a coin at every fork and let chance decide. At every fork, the group has only a 50 percent chance of getting it right. Let us call this decision procedure “random.”

A second option is to let one person make the decisions for the group at every fork. This person could claim the role because people believe she has a special connection with an all-knowing God, or because she can plausibly claim to be an expert in maze solving, or for whatever other reason. The group can let that person self-appoint or actively elect her. We can generally label the rule of such a person a dictatorship. Even if the dictator is initially elected, her power is not subjected to the classical accountability mechanisms of democratic representation, such as the challenge of repeated elections after the first (per Manin 1997).

The group can also choose to let a small council of people make decisions for the group at every fork. Even if the members of the council did not impose themselves by force or cunning but were chosen by the rest as smart and capable of making decisions on behalf of the group, we will label this council of experts an “oligarchy.” Here again, the absence of a system of periodic elections and other accountability mechanisms makes it impossible to see this option as even remotely representative, let alone democratic.

A fourth option is for the group to choose at every fork to decide as a group, through deliberation followed by majority rule. If the group is too large, representatives (whether chosen by elections or a lottery organized at regular intervals, for example, at every new fork in the road) can make decisions on its behalf. We will label this option “democracy.” When the whole group is directly involved in the decision, we are dealing with direct democracy. When only a subset of representatives is directly involved, we are dealing with indirect or representative democracy. For now, we will assume that representative democracy is not fundamentally different from direct democracy.

The choice is thus between a random procedure, a dictator, an oligarchy, and a direct or an indirect democracy. This book argues that, for the

⁵I assume for the sake of simplicity that choices are binary or can be reformulated as a long series of binary choices. I address some of the difficulties with that assumption (including the issue of agenda manipulation) in chapter 7.

purpose of getting out of the maze as a group, the democratic alternative, whether in a direct or indirect form, is not just better than random but better than the idealized dictator and the oligarchy. If there were only one fork and thus only one choice to make, letting a dictator decide on that one occasion might be as good an option as choosing as a group, since if the group were lucky, it would pick as a decision maker the one person who actually knew the answer on that particular occasion. As the problem poses a *long series* of choices to be made, however, it becomes very unlikely that the person elected as a dictator at the beginning of the journey, however smart and informed she might be, will know the right answer at every new fork in the road. It is more likely that someone else in the group will know the answer, or only the group as a whole will know the answer. It is smarter, therefore, to keep as many people as possible involved in the decision every time the group arrives at a new fork.

The comparison of democracy with the small council might seem trickier. If the group elects its brightest members at the beginning and asks them to make decisions on its behalf, it seems that it would have a better chance to get out of the maze than if it involves everyone, including the not-so-smart people. As we will see, however, group intelligence is only partly a function of individual intelligence. It is also a function of cognitive diversity—roughly, the existence of different ways of seeing the world—which is a group property. If the brightest individuals form a small group in which everyone happens to be nearsighted (because brightness is correlated with extensive reading, for example), they will not as a group have as much cognitive diversity as a larger group. Consequently they will miss the clues written in small fonts on the walls, which perhaps the not-so-bright people with good vision would be able to pick up.

A more inclusive decision-making process, or one that would renew the pool of decision makers through regular elections, would in that case produce a better decision. The advantage of democracy over a dictatorship and even a small group of experts is strengthened the longer and the more complicated the maze is. There is practically no chance that one single person could resolve a long series of difficult puzzles. There is a better chance that a small group of experts could, but the probability is still small. The way to maximize the probability of getting all the answers right and finding the way out of the maze is to include everyone, whether directly or indirectly.

Notice that assuming that the people elect the dictator or the group of oligarchs stacks the deck against democracy in the first place. If the people are generally competent at making political decisions, they should also be fairly competent at choosing their leaders, which means that an elected dictatorship or an elected oligarchy would be the most effective forms of these regimes. In practice, dictatorships and oligarchies are usually

imposed from the top down, which rarely ensures that the individuals who end up as dictators or oligarchs are the best that they can be. They are more likely to be violent and power hungry than to be effective at pursuing the good of the country. What I aim to show is that democracy is epistemically superior to the rule of the few not just when the few are traditionally defined (as unelected and generally self-appointed oligarchs) but in fact even when the few are elected.

The metaphor of the maze has its limits, to which I will return in the conclusion. Assuming for now that I succeed in rendering plausible the claim that democracy does at least better than a random decision procedure and also better than alternative nondemocratic procedures, one may ask, first, What is the gain in terms of the normative justification for democracy? The answer is: a lot. At a minimum, to the extent that the goal of a justification for democracy is simply to establish the value of democracy as an instrumentally desirable regime, an argument showing that democracy has strong epistemic properties can only be welcome. At a maximum, the argument can reinforce, if not altogether establish, the legitimacy of democratic authority.

Second, if the goal of a justification happens to overlap with the task of legitimation of democracy as a normatively desirable regime—if, in other words, we do not draw a strict distinction between justification and legitimation—then the epistemic case for democracy may prove crucial to establishing the normative authority of democracy. David Estlund has opened this path by arguing that the normative concept of democratic authority includes an epistemic dimension (1997; 2007). Against consent theorists who believe that legitimacy is strictly a matter of the consent of the people and has nothing to do with the reasons that may or may not justify a regime, Estlund's "epistemic proceduralism" reintroduces instrumental considerations in the legitimation of democracy. In his view, democracy would have no normative authority, that is, no right to claim obedience to its decisions, if we did not assume that it met a minimal epistemic threshold, which he sets at "better than random." Estlund, however, is not really interested in providing the actual proof that democracy can be expected to meet that minimal requirement. If successful, the case presented in this book will provide such a proof.

Finally, to the extent that the book aims to prove an even stronger claim—namely that democracy epistemically outperforms nondemocratic rules—its contribution is to offer an argument for democracy (and, possibly, democratic authority) even to people who do not share the Western faith in the right of people to govern themselves when this right is entirely disconnected from the question of their capacity for it. One could imagine, for example, a person who subscribes to the idea that political

authority must include an epistemic dimension but does not think that nondemocratic regimes are by definition disqualified from claims to legitimacy and normative authority. For such a person, an important question is the following: assuming that all regimes meet a minimal threshold of epistemic competence, what is it about democracy that makes it more normatively authoritative or legitimate than an equally competent oligarchy of the wise? There is a readily available argument for such a skeptical reader: democracy is simply a smarter regime than the rest.

While the contribution of this book should thus be clear, let me also emphasize its originality, which has to do with the connection drawn between inclusive decision making and the role of cognitive diversity in the emergence of democratic reason. All sorts of democrats have argued that a diversity of points of view, and even active and passionate dissent, are healthy for a democracy. From John Stuart Mill's defense of social gadflies to Cass Sunstein's celebration of dissent (Sunstein 2003), the case has been abundantly made that the existence of heterodox thinkers—people who think differently, including in violation of the group's most fundamental norms—spur the body politic away from complacent rehearsal of dead dogmas and toward more creative thinking.

This case for the cultivation of different viewpoints suggests tempering any form of government with liberal rights, including, crucially, freedom of expression. It does nothing, however, to support more-inclusive decision making as such. Although James Surowiecki (2004) has made the case for more-inclusive decision making on the basis of the statistical properties of large numbers and the Condorcet Jury Theorem, he says little about cognitive diversity *per se*. He also tends to focus only on the purely aggregative side of collective intelligence, dismissing its deliberative aspects as counterproductive and conducive to polarization. What I propose in this book is an argument that explicitly connects the epistemic properties of a liberal society and those of democratic decision procedure. I argue that in an open liberal society, it is simply more likely that a larger group of decision makers will be more cognitively diverse, and therefore smarter, than a smaller group. I thus attribute the epistemic superiority of democracy not only to the sheer number of decision makers, but also to the qualitative differences that, in liberal open conditions, this great number of decision makers is likely to bring with it. For all its simplicity, this insight is, I believe, new and important.

The epistemic argument for democracy based on the centrality of cognitive diversity that I defend in this book has at least three characteristics: it is maximal rather than minimal; it is probabilistic; it is theoretical *a priori* rather than empirical (and yet can be translated into an empirical and thus falsifiable claim).

On the minimal version of the epistemic argument for democracy, democracy is at least as good as, and occasionally better than, a random decision procedure at making decisions, although it can be inferior to rule by the wise few or the lone genius. On the maximal version of the epistemic argument for democracy, democracy is at least as good as, and occasionally better than, any alternative decision rule. Between those two extremes lie many intermediary positions. By choosing to defend the maximal version, I take a risk vis-a-vis other epistemic democrats, who simply compare the possible epistemic properties of some democratic procedures with those of other democratic procedures, such as deliberation versus majority rule, or compare democracy with a random procedure.

Note that I am not proposing a full-blown theory of democratic authority. I am interested here only in the epistemic side of a justification for democracy, which is in my view a necessary component of such a justification, although perhaps not a sufficient one. The epistemic argument for democracy that I defend is thus freestanding and conceptually independent from arguments for democracy that emphasize, for example, the fairness of the democratic decision process. Whether I succeed or fail in arguing for the maximal epistemic claim, democracy would still be an attractive form of government, as long as I (or others) succeed in establishing the minimal claim that it does at least as well and occasionally better than a random procedure. Above this threshold, classical pure proceduralist arguments can apply.⁶ As long as democracy does better than random—that is, it performs minimally well—the fact that it embodies our ideals of equality and autonomy arguably would make it a worthy choice.

Another way to say this is to insist that the maximal epistemic argument for democracy presented here need not be the primary, let alone the only, argument for democracy. In the history of thought, democracy has generally been primarily justified on nonepistemic grounds, through appeal to notions of justice, equality, fairness, or consent. In earlier ages, an epistemic argument was doomed to failure in the face of deeply entrenched prejudices against the masses and the lack of an empirical record demonstrating how well democracies can do. This historical legacy has left us with a set of reasons for valuing democracy that are largely nonepistemic.

However bold it may seem, the epistemic argument defended here is—and this is its second characteristic—only probabilistic. The claim that democracy is overall the smartest method for making group decisions

⁶By “pure proceduralism” here, I mean the opposite of pure instrumentalism, namely a position that considers that the value of democracy lies in some intrinsic properties of its procedures, rather than in the quality of its outcomes.

does not exclude the possibility that some democratic decisions will be mistaken, nor does it exclude the possibility that a particular democracy will do worse than a particular oligarchy. Some of the time, the argument implicitly acknowledges, a democratic decision will prove inferior to one that would have been imposed by a dictator or a group of aristocrats or experts. Some of the time, a democracy will fail where a dictatorship or an oligarchy would have succeeded. On average, however, and in the long run, the claim is that democracy is a safer bet than a dictatorship or even an aristocracy.

The third characteristic of the epistemic claim presented in this book is that it is theoretical and *a priori* rather than empirical. I thus rely on models and theorems to support my case for democracy, rather than on case studies or empirical evidence. The probabilities I am thus considering in this book are theoretical or *a priori* probabilities, not frequencies (that is, empirically observed regularities). Where I do resort to empirical examples, they are meant only to support the plausibility of the theoretical claim, not establish it. This theoretical and *a priori* bent of the argument stems from the nature of the object itself: I am interested primarily in the *ideal* of democratic decision making. The reason is that while it may seem more important to focus on ways to improve the reality of democracy, this practical task presupposes first a clearer understanding of the ideal and why it is an ideal in the first place.

Although it is *a priori*, the epistemic claim put forward in this book can, I believe, be translated into an empirical and thus potentially falsifiable claim, provided the theoretical assumptions are shown to have a real-life equivalent. Wherever possible, I try to show that the assumptions behind various theorems (for example the Diversity Trumps Ability Theorem or the Condorcet Jury Theorem) plausibly translate to the real world. The theory thus allows me to venture a prediction: decisions about public affairs made by the many should be probabilistically (this time in the sense of “statistically”) superior to decisions made by the few. While I do not myself build the empirical case to support this prediction, I mention existing supporting evidence whenever appropriate and trust that more could ultimately be found, whether in the form of qualitative or quantitative studies, case studies of specific democracies (as in Ober 2010), or large-N studies of groups of democracies. Of course, a lot hinges on what the empirical standard of “intelligent political decisions” is taken to be. Aside from the conceptual problems attached to the idea of such a procedure-independent standard of rightness, which I address in chapter 8, construing an empirical equivalent raises technical and methodological challenges that I cannot touch on here.